

Project 1: Linux Fundamentals Lab

Purpose

Learn basic Linux navigation and file management.

Commands to Learn

Pwd	Shows current directory path		Nano	Opens a simple word editor
Ls	Shows files and folders in current location		Echo	Repeats text, can be used to add to txt files
Cd	Moves between directories		Ctrl+o	Save
Mkdir	Creates a directory		Ctrl+x	Close
Touch	Creates a file		Cat	Reads the file

Practice:

- Creating files
- Moving files
- Renaming files
- Deleting files
- Viewing file contents

Questions You Should Be Able To Answer

What does pwd do?

What is the difference between cp and mv?

What does cat do?

What directory am I currently in?

How do I create a file?

How do I move a file?

How do I search for a file?

What is the difference between copying and moving?

My Findings:

I installed Kali Linux and Oracle Virtual box to set up my own lab environment.

I opened the terminal using ctrl+T and decided I first needed to know where I was. I used PWD (print working directory) to determine my location. I was in /home/asignia

Now that I knew where I was I wanted to see what was in this location. To do that I used ls and found the following:

```
(asignia@kali)-[~]  
$ ls  
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

So I needed to create some more folders to explore, move, copy, etc.

to do this I first created a directory using mkdir. I created a directory called CyberLab and verified it was there using ls.

```
(asignia@kali)-[~]  
$ mkdir CyberLab  
  
(asignia@kali)-[~]  
$ ls  
CyberLab  Documents  Music  Public  Videos  
Desktop  Downloads  Pictures  Templates
```

I decided to enter the new directory and take a look around.

```
(asignia@kali)-[~]  
$ cd CyberLab  
  
(asignia@kali)-[~/CyberLab]  
$ ls  
  
(asignia@kali)-[~/CyberLab]  
$ pwd  
/home/asignia/CyberLab
```

Ok, now that im here, lets create some files.

I used touch to create a file called 'notes.txt'

```
(asignia@kali)-[~/CyberLab]  
$ touch notes.txt  
  
(asignia@kali)-[~/CyberLab]  
$ ls  
notes.txt
```

Now I did need to look up how to add to that initial text document. I knew how to create one and find it but not how to give it any purpose. So after a quick search using man man

```
(asignia@kali)-[~/CyberLab]
$ man
What manual page do you want?
For example, try 'man man'.

(asignia@kali)-[~/CyberLab]
$ man man
```

This was so that I could see a list of commands available to me, unfortunately it does not specify a command for editing a text file. My next step in my search was good ol' google. And it gave me two methods to try. The first was 'echo' and the second was 'nano' I will show both below.

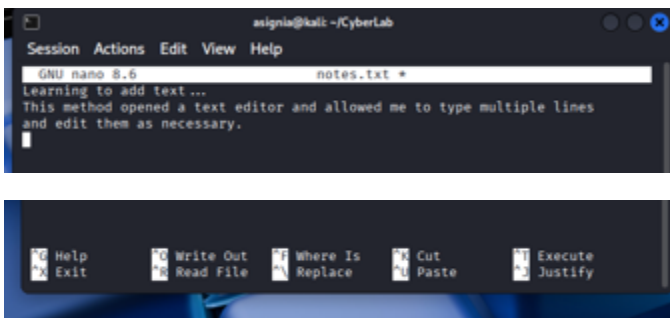
I also checked to make sure that the text I wanted to add was, in fact, visible on the file by using 'cat'.

```
(asignia@kali)-[~/CyberLab]
$ echo "Learning to add text" > notes.txt

(asignia@kali)-[~/CyberLab]
$ cat notes.txt
Learning to add text
```

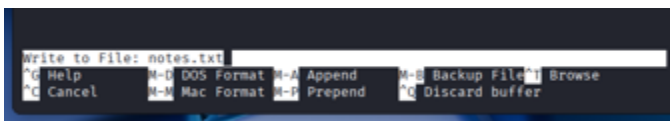
Awesome! That method worked, now let's try the other one.

The next command was 'nano'. This method opened up a simple text editor that I was able to use more easily and edit as needed. However this method did take a bit of practice as well. You can not click around the text box and the line will simply keep going, adding a scrollbar, unless you press 'enter'. Another useful feature of this method was that there was a list of commands at the bottom.

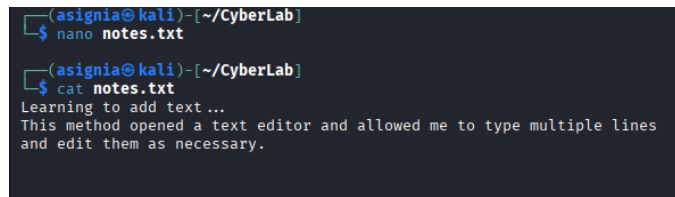


When I was finished adding my text I went ahead and used that handy key to 'write out', or save my document using ^O.. but wait, I tried that command and it just wrote it into my document. I had to look up how to use those commands instead of just write them.

Ah-ha! The ^ is supposed to be a stand in for 'ctrl' nice! Well shoot, now that I can save I have another new option.. I again went to google to ask about these additional options.



It explained that those additional options are there if you would like to change the file format. Since I did not wish to do this, I simply hit enter and then closed the file.

A terminal window with a dark background. The prompt is (asignia@kali) - [~/CyberLab]. The first command is nano notes.txt. The second command is cat notes.txt, followed by the output: Learning to add text ... This method opened a text editor and allowed me to type multiple lines and edit them as necessary.

```
(asignia@kali) - [~/CyberLab]
$ nano notes.txt

(asignia@kali) - [~/CyberLab]
$ cat notes.txt
Learning to add text ...
This method opened a text editor and allowed me to type multiple lines
and edit them as necessary.
```

Voila!

Day 2-

cp	Copy		>>	append
Mv	Move		>	Overwrite
rm	Deletes files and directories			

Lessons Learned

- Linux uses commands instead of graphical interfaces for many administrative tasks.
- Documentation and research are essential parts of learning unfamiliar systems.
- Multiple methods can accomplish the same task (echo vs nano).
- Understanding command syntax is just as important as memorizing commands.